

Name:

Stud.ID:

# Vision and Cognitive Systems

A.Y. 2021/2022

Midterm Exam - December 6, 2021

## Exam Instructions

- Write your *name* and *Stud.ID (matricola)* on **every** page.
- You have just one page (one side) to answer a question.
- You will have **exactly** 90 min to complete your exam.

**Question 1 (2 points)**

Describe the main components of a supervised machine learning model (e.g. linear regression), i.e. the hypothesis function, the cost function and how the learning works.

**Question 2 (3 points)**

What's the purpose of image filtering? You should give a proper definition of filtering and clearly define the mathematical tools used in order to define any filter. Finally, report an example of a real filter and describe what it can do.

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### Question 3 (3 points)

bla bla bla

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### Question 4 (2 points)

bla bla bla